MIL-W-6873B
6 SEPTEMBER 1968

Superseding
MIL-W-6873A
19 January 1959

MILITARY SPECIFICATION

WELDING; FLASH, CARBON AND ALLOY STEEL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers procedures for flash welding alloy steels and steels of medium and low carbon content.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-I-6868	Inspection Process, Magnetic Particle
MIL-H-6875	Heat Treatment of Steels (Aircraft Practice) Process for

STANDARDS

Federal

FED. TEST METHOD STD. NO. 151	Metals; Test Methods

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. EQUIPMENT

3.1 Requirements.- Means shall be provided for automatically controlling the feed of the joint during flashing, the rates and distance of travel of the

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sections to be welded, and for controlling the secondary voltage and current magnitudes and timing, and the time of current cutoff.

3.1.1 The equipment shall be capable of bringing the two sections to be welded into contact, initiating the flow of welding current and forming a "flashing" action between the adjacent pieces of metal. As the metal on the surfaces is burned or flashed away, the movable platen shall advance and maintain the gap within appropriate limits such that flashing is maintained until the mating surfaces and adjacent areas have attained a degree of plasticity suitable for welding. The surfaces shall be rapidly brought together with sufficient force to close all craters formed during flashing, expel molten metal and substantially all oxides and impurities from the weld proper, upset the adjacent metal, and produce welds of acceptable quality.

3.2 Operation.- Once the process has been initiated by the operator, the equipment shall follow the sequence of operations automatically and complete the weld. The equipment shall consistently produce welds which meet the quality standards specified herein. Manual flashing not to exceed 15 percent of the total flashing travel will be permitted.

3.3 Electrodes.- The electrodes shall be of adequate size, and shall be so fitted to the surfaces of the parts to be welded that the necessary current will be uniformly distributed about the contact zone, and shall be conducted from the electrode to the work without damage owing to localized overheating.

3.4 Control of machine variables.- The equipment shall be set up by experienced operators under the supervision of a welding engineer, or other authorized personnel.

3.4.1 Welding schedules.- Complete flash welding procedure schedules shall be established as specified herein for the joints to be welded in production and approved by the authorized inspector before a machine may be used in the welding of items to be furnished on Government contract.

4. MATERIALS AND PREPARATION

4.1 Materials.- Plain carbon and alloy steels of not more than 0.50 percent nominal carbon content may be welded, provided all requirements of this specification are met.

4.1.1 Cleaning.- The surfaces of the parts contacting the electrodes, and the surfaces at the fusion zones, shall be clean and free from oxides, paint, grease, dirt, or foreign matter which would interfere with the flow of current or fusion of the metal. Chemical cleaning or grit blasting are preferred. Sandblasting may be employed provided the surfaces are treated to remove embedded sand particles prior to welding.

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4.1.2 Joint preparation.- The mating parts shall be aligned in such a manner that the heat generated by the flow of current will be uniformly distributed over the section. The details of joint preparation shall be determined by the contractor and shall become a part of the welding schedule for the respective joint. Recommended practices are shown on figure 1.

4.1.3 Limitation on tubing.- Tubing possessing a ratio of outside diameter to wall thickness of greater than 30:1 shall not be flash-welded unless the equipment is capable of meeting the more rigid alignment tolerances for 35:1 and 40:1 ratios specified in 5.2.1.

5. QUALITY STANDARDS

5.1 Flash-welded joints shall be sound and free from injurious defects, such as cracks, inclusions, die burns, and porosity which affect the strength or serviceability of the part. Welds with suspected discontinuities will be acceptable only when tests indicate that all requirements in this specification are met.

5.1.1 Flash removal.- A sufficient amount of the extruded or upset metal shall be removed from all weld areas in order that no defective or unsound material remains. A weld shall be rejected if it is necessary to machine below the level of the surface of the parent metal to remove surface defects. Unless otherwise specified, the removal of the internal flash from joints in hollow shapes will not be required.

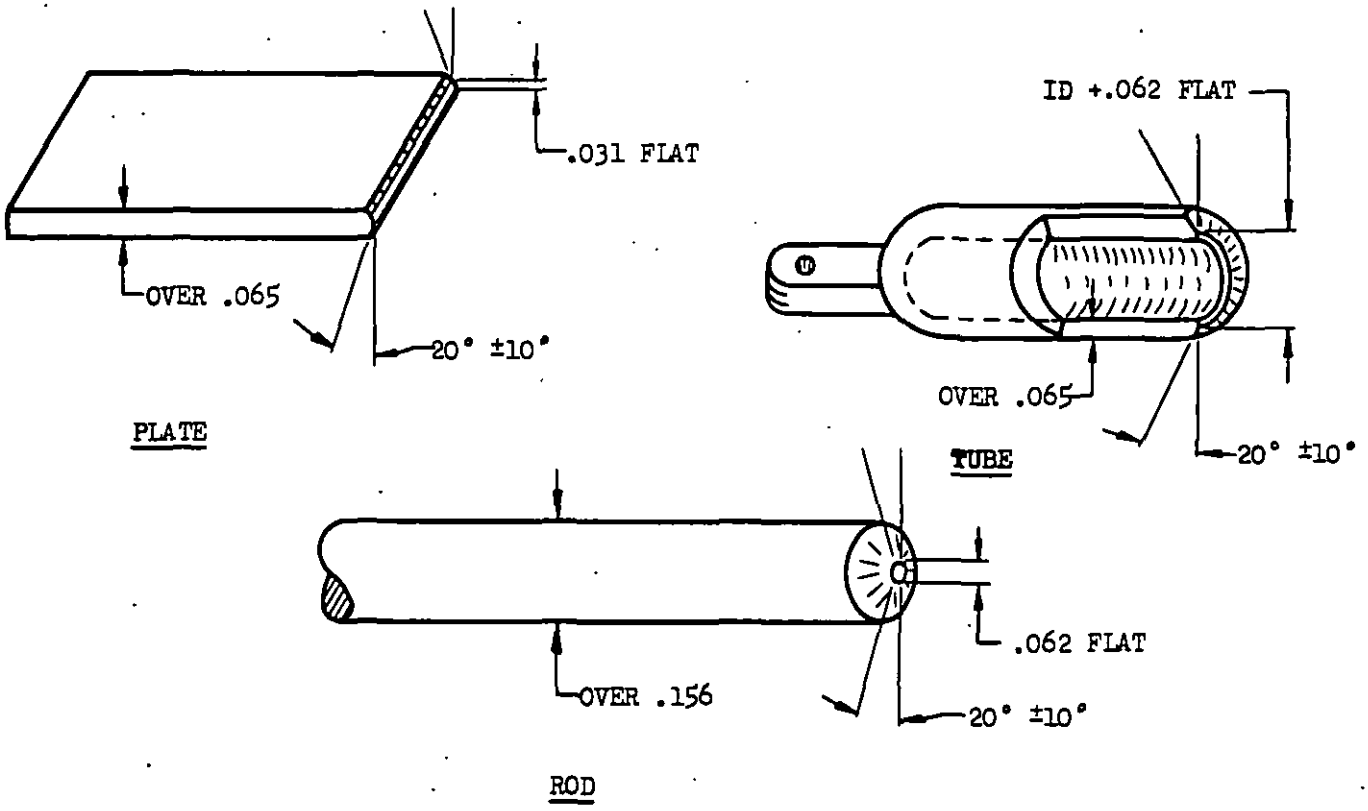
5.2 Alignment tolerances.-

5.2.1 Sheet and tubular joints.- For sheet thickness or wall thickness of tubing 0.080 inch and less, the misalignment of thickness shall not exceed 0.005 inch or 8 percent of the nominal thickness, whichever is greater. For sheet or wall thickness greater than 0.080 inch, the misalignment of thickness shall not exceed 10 percent of the nominal section thickness up to a maximum of 0.030 inch, except for the following two higher ratios where a smaller misalignment is required:

O.D. wall ratio	Misalignment tolerance Percent maximum
Over 30:1 to 35:1	6
Over 35:1 to 40:1	4

Misalignment shall be measured at, or near, the plane of the weld before the flash is removed.

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NOTE: a. THE ENDS OF PARTS TO BE WELDED SHALL BE OUT SQUARE WITHIN PLUS OR MINUS 1/2 A DEGREE.

b. WHEN THE NOMINAL STOCK THICKNESS EXCEEDS THE VALUES GIVEN IN THIS FIGURE, CHAMFER ONE OF THE PARTS AS SHOWN. IN THE EVENT IDENTICAL PARTS ARE TO BE WELDED TO EACH OTHER RESULTING IN EACH BEING CHAMFERED, CHANGE THE ANGLE TO 10 DEGREES.

UNLESS OTHERWISE SPECIFIED, DIMENSIONS IN INCHES.

FIGURE 1. Recommended end preparation for flash butt welding of tubing, flat sheets, and bar

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5.2.2 Bars.- The misalignment of the surfaces in joints between bars and rods shall not exceed 5 percent of the diameter or thickness when measured at, or near, the plane of the weld, before the flash is removed.

5.2.3 Angular misalignment.- The angular misalignment of the welded joints shall not exceed 0.002 inch per inch of length.

6. CERTIFICATION OF WELDING SCHEDULES

6.1 General.- The welding schedule shall be established by the contractor for each joint differing in material, area or contour from joints previously welded in accordance with approved schedules, or for each welding machine over an appropriate range of cross-sectional areas, or wall thicknesses, or both.

6.2 Welding schedule.- A complete welding schedule shall include the preparation of the parts to be welded, choice of electrodes and fittings, and appropriate machine control devices and settings to provide the proper sequence of events. The record of a welding schedule shall include the appropriate items of figure 2.

6.3 Establishing welding schedules.-

6.3.1 General.- The three initial welding schedules established by the contractor for a given machine shall represent joints of the greatest and least cross-sectional areas to be welded in production plus one joint of intermediate areas. To determine the acceptability of the settings and the consistency of resulting welds, 15 or more joints shall be welded at each of the three points. Each joint shall be inspected as specified in 7.4.1 and 7.4.2 and tested as specified in 6.4.2. The test data will be submitted to the authorized inspector with request for approval of the welding schedules represented. Assemblies welded with the respective machine shall not be acceptable as items furnished on a Government contract until the suitability of the welding schedules has been demonstrated as specified herein. Weld schedules shall not be changed during a production run, except as provided in 6.3.4.

6.3.2 Plotting of control settings.- Graphs of the following items or their equivalents shall be plotted by the contractor and used as an appropriate guide in establishing welding schedules for other joints as required:

(a) Machine settings (such as upset, final die opening, material loss, etc.) versus wall thickness or cross-sectional area. (See figure 3 - specimen graph.)

(b) Flashing time versus wall thickness. (See figure 4 - specimen graph.)

(c) Voltage versus cross-sectional area. (See figure 5 - specimen graph.)

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Manufacturer of machine_____
Type and serial number_____
Capacity of machine_____
Transformer tap setting_____
Transformer turns ratio_____
Secondary voltage (approx.)_____
Flashing time (seconds)_____
Beginning of upset to current cutoff_____
Burnoff (inches)_____
Upset travel (inches)_____
Platen travel control cam (when used)_____
Area of electrode contact areas_____
Work locaters employed_____
Material and surface condition_____
Details of joint preparation:
(Show sketch and dimensions)

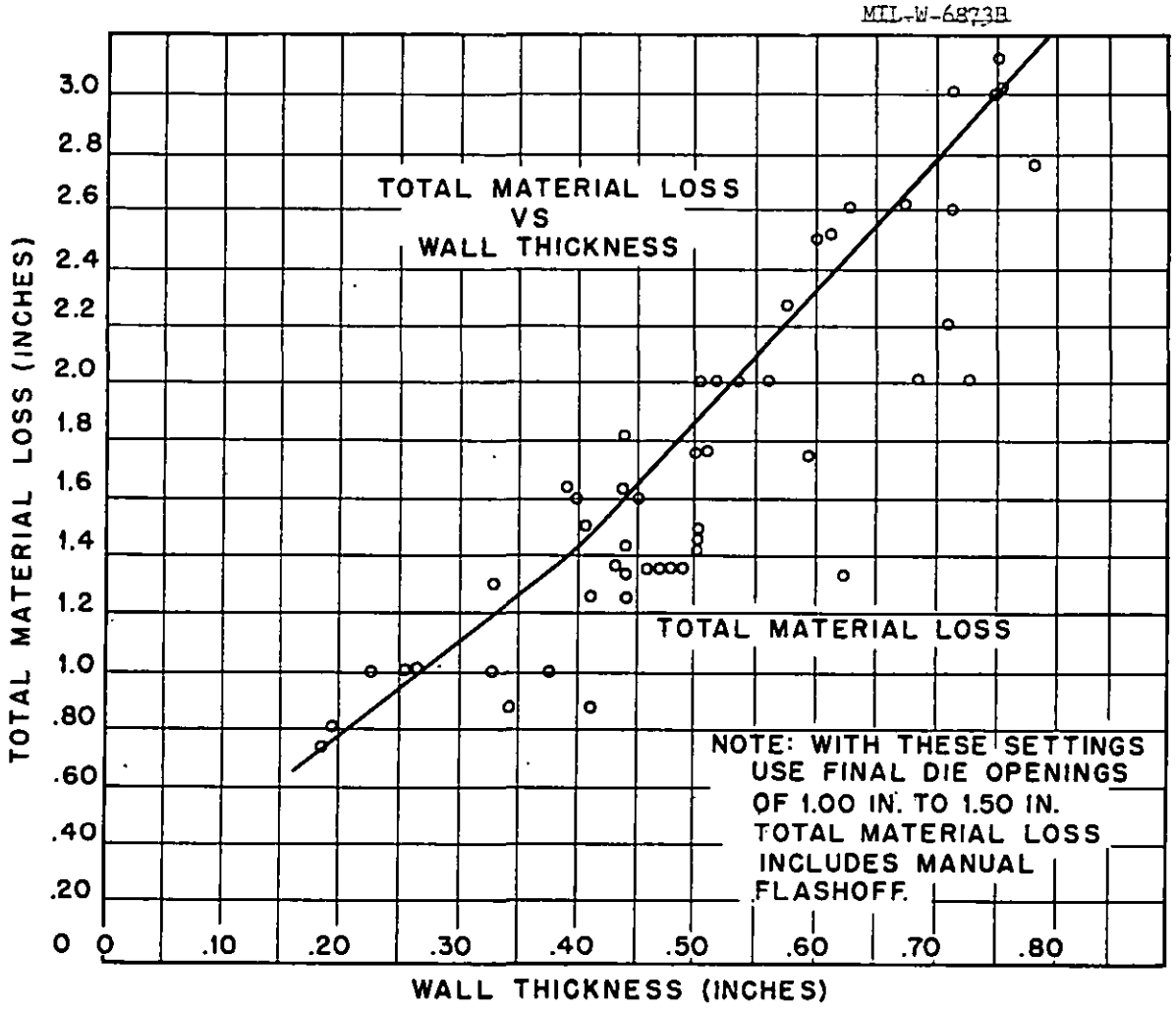
* _____
Signature (Welding Engr.)

* _____
Signature (Authorized Inspector)

* NOTE: The supervisory welding engineer or authorized alternates, and the authorized inspector shall sign the welding schedule form.

FIGURE 2. Record of welding schedule

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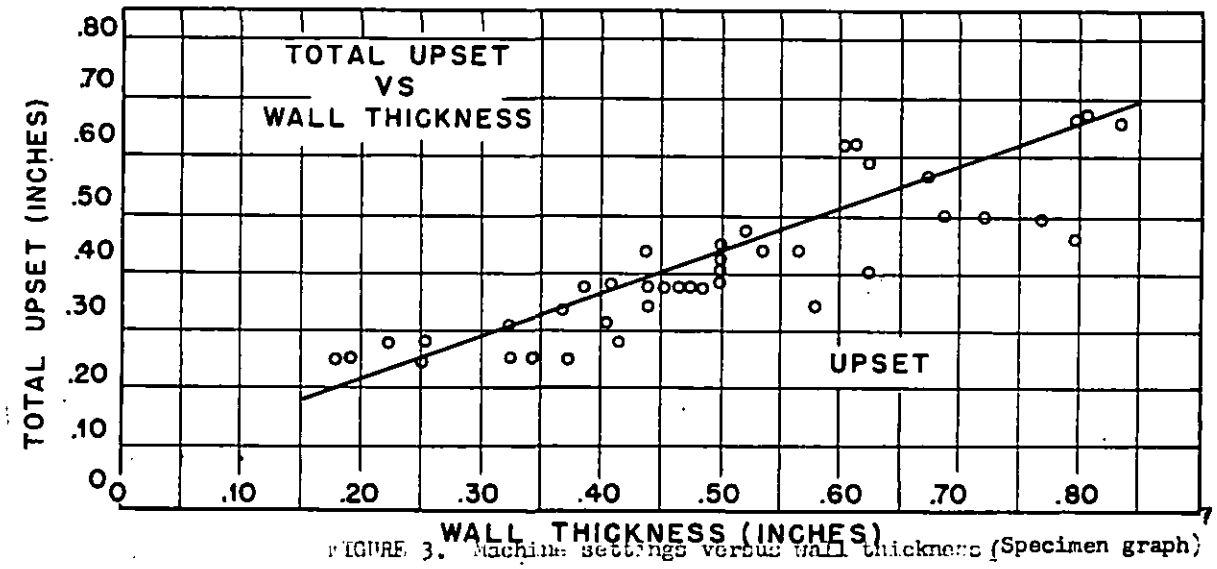


FIGURE 3. Machine settings versus wall thickness (Specimen graph)

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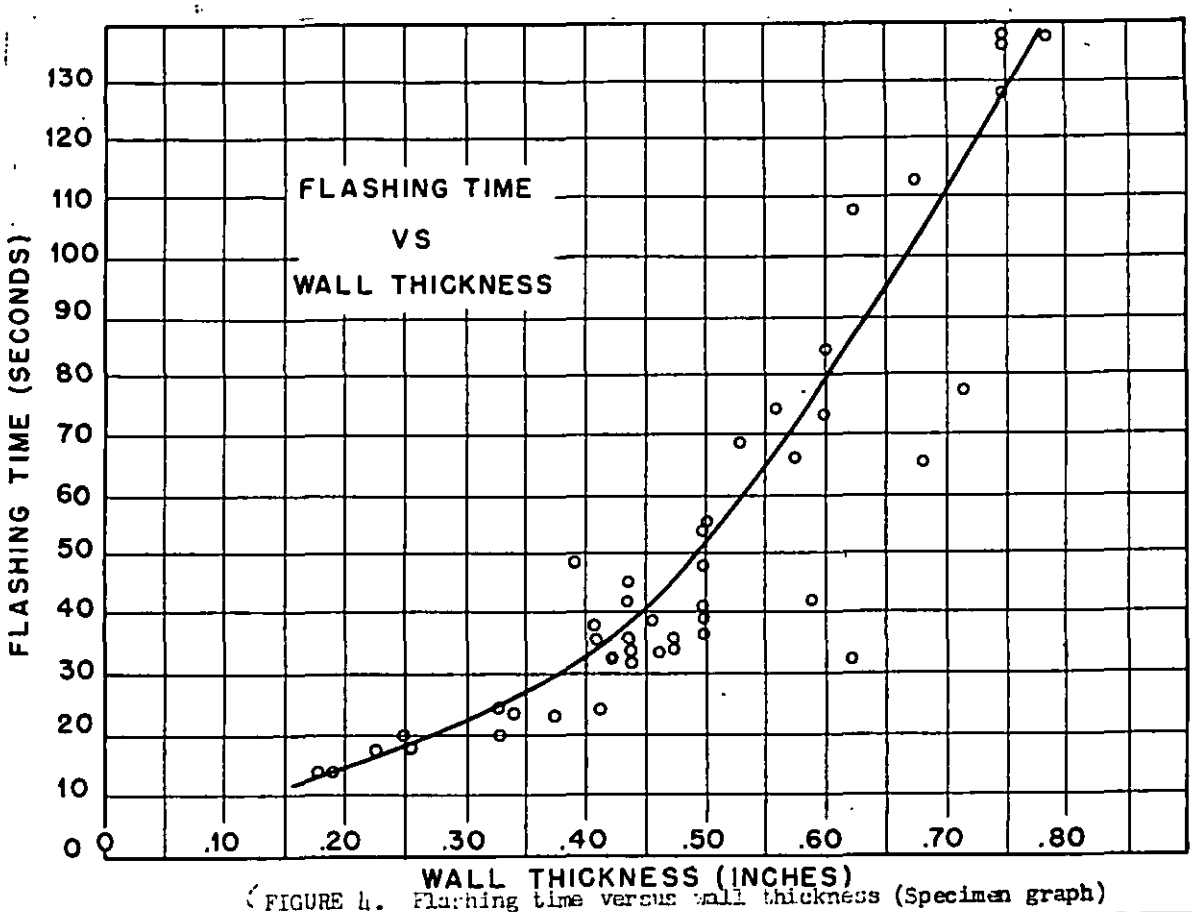


FIGURE 4. Flashing time versus wall thickness (Specimen graph)

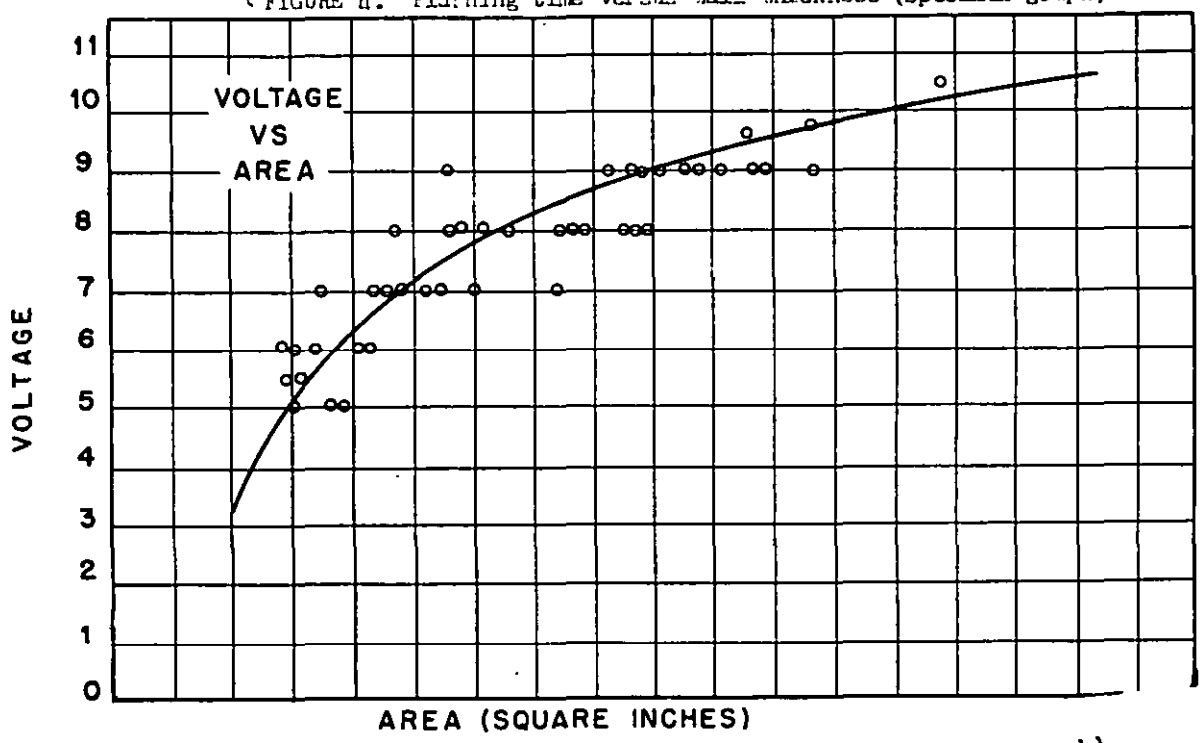


FIGURE 5. Voltage versus cross-sectional area (Specimen graph)

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6.3.3 Alternate method.- When welding data for a machine have been accumulated by the contractor prior to certification for use under Government contract, and the settings are appropriate and records complete, the data may be used as an alternate basis for establishing the required graphs. In such cases, the suitability of all welding schedules shall be demonstrated in accordance with the procedure specified in 6.3.4.

6.3.4 Additional schedules or modification of existing schedules.- To weld parts with cross-sectional areas other than those specifically checked above, or to accomplish minor modification of existing schedules as may be necessary, the contractor shall demonstrate that the settings for these secondary points are acceptable on five sample parts or control specimens. If unsatisfactory results are obtained for a secondary point, a sample of 15 shall be run to check the point, after suitable adjustments in settings have been made. Specific approval of each additional schedule will not be required, but complete records shall be made readily available to operating personnel and authorized inspection personnel for reference at any time.

6.3.5 Development of graphs.- Additional settings shall be plotted as available and the graphs adjusted as experience indicates. When sufficient data have been accumulated, a band will be developed for each machine within which acceptable welds may be obtained.

6.3.6 Individual part certification.- When one, or a limited number of joint types are to be welded in production and complete qualification of the machine is not considered necessary or desirable, the contractor may establish individual schedules on the basis of 15 test specimens.

6.3.7 Production run.- Unless otherwise specified, one setting of the machine shall satisfactorily weld the maximum and minimum cross-sectional areas scheduled during a production run. No change in setting shall be made during a run without the approval of the authorized inspector.

6.4 Certification test specimens.- Either production parts or simulated parts may be used in welding certification test specimens. The parts shall be prepared under production conditions. The mating surfaces to be joined and the electrode contact areas shall be identical with the production part in cross-sectional area, contour, and finish. The joints shall be welded in accordance with the proposed welding procedure and with the requirements specified herein.

6.4.1 Heat treatment of specimens.-

6.4.1.1 Joints for service as welded.- Welded specimens representing parts made of steel in the normalized or the "as received" condition to be used as welded and without further heat treatment may be stress relieved prior to testing.

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6.4.1.2 Joints to be heat treated prior to service.- Welded specimens representing parts to be heat treated to higher strength levels shall be heat treated to not less than the design strength level in accordance with MIL-H-6875.

6.4.2 Type of specimens.- Specimens shall be tested in full section, whenever practicable. Where test specimens are too large to be tested in full section, tensile coupons conforming to an appropriate type as defined under Method 211 of Federal Test Method Standard No. 151 may be cut from locations, such that the weld zone is included in the test section. Tensile coupons from tubular specimens may be prepared as indicated in figure 6. The aggregate weld area of the coupons shall not be less than 50 percent of the area of the weld joint represented, except when a lower percentage is allowed under the following conditions: In the initial certification tests, the aggregate weld area of the coupons may be reduced to 25 percent when nondestructive inspection procedures are used to examine the welded joints and when the suitability of the nondestructive inspection is established by the tensile tests.

7. TEST PROCEDURES

7.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

7.2 Production control.- All specimens required herein for checking production flash welding shall be welded in accordance with the established schedule.

7.3 Sampling.-

7.3.1 Destructive sample.- Each time the machine is set up for a production run, the setup shall be checked by welding and testing three parts or simulated production parts as specified in 7.4.4, except that samples may be tested in the "as-welded" condition and will be considered acceptable provided that failure is outside the weld zones; destructive tests of large weldments, on which testing in full section is not practical, will not be required, provided every production joint is proof-tested at the loading conditions specified in 7.4.5. Alternate nondestructive inspection procedures may be used provided the suitability of the procedures has been established by destructive tests.

7.3.2 Proof-testing sample.- Unless greater frequency of sampling is specified, production lots of critical and structural parts shall be sampled for proof testing as follows:

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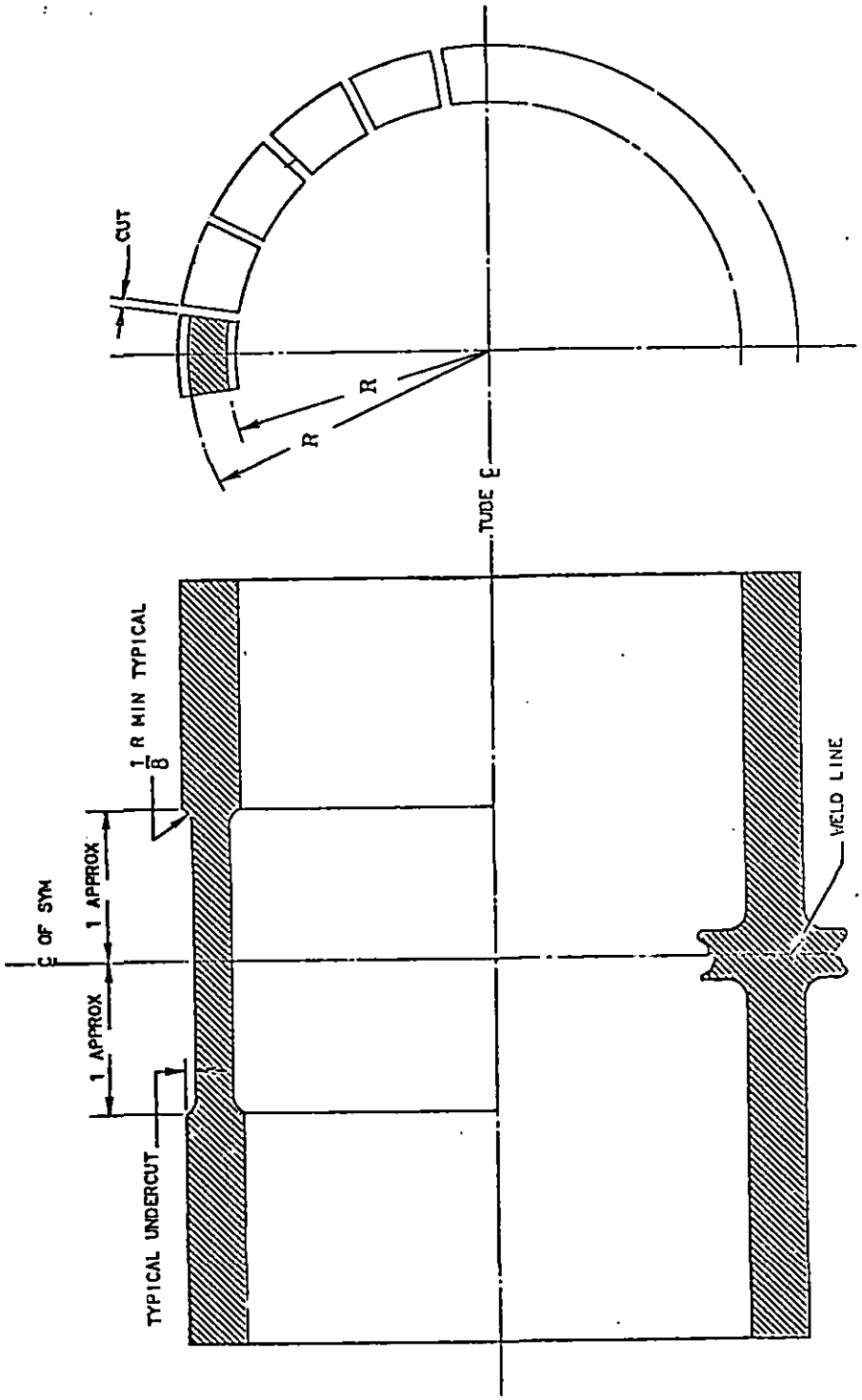


FIGURE 6. Alternate configuration for friction gripped tensile coupon

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Lot size	Sample size
Less than 31	All
31 - 50	30
51 - 100	42
101 - 200	55
201 - 400	60
401 - 800	65
801 - 4,000	70
4,000 - 10,000	75

7.4 Inspection methods.-

7.4.1 Visual examination.- Each flash-welded joint shall be examined for cracks, die burns, telescoping, and other indications of unsound welding before and after the flash is removed.

7.4.2 Magnetic inspection.- All production welds in magnetic materials shall be inspected by magnetic particle inspection in accordance with MIL-I-6868.

7.4.3 Destructive tensile test.- Destructive tensile test shall be performed in accordance with the requirement specified in 7.4.4. Other methods of destructive tests such as bend tests, impact tests or short-time fatigue tests, as sensitive as the tensile test to variation in weld quality may be used when approved by the procuring activity.

7.4.4 Tensile tests.- The whole weld test specimen shall be subjected to tensile tests in accordance with Federal Test Method Standard No. 151. Where test specimens are too large to be tested in whole, they may be cut into tensile coupons prior to testing in accordance with Federal Test Method Standard No. 151, except that reduction in size of the test section of the specimens will not be required. Specimens may be stress-relieved or normalized to facilitate machining, but shall be heat treated as specified herein prior to testing. The entire test specimen, except that removed by machining, shall be tested when cut into coupons. The ultimate strength of the parent metal will be determined from the average of not less than three tensile tests. Specimens of welded joints having failures completely outside of the weld may be substituted for tensile specimens of the parent metal.

7.4.4.1 The results of the above tensile tests shall furnish evidence that the parent material has been heat treated as specified above.

7.4.5 Proof tests.- All specimens of the production sample selected as specified in 7.3.2 shall be proof-tested in tension or bending. Wherever possible, the proof load shall correspond to at least 67 percent of the ultimate strength specified for the material. If the part has a particular region which is weaker than the flash-welded joint, the proof testing load shall be equal to at least 67 percent of the ultimate strength of the weakest region of the part. Parts which require heat treatment after welding shall be proof-tested after heat treatment.

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7.5 Test requirements.-

7.5.1 Whole specimens.- The results of the tests on whole specimens shall conform to the following:

(a) No specimen tested in the "as-welded" condition shall fail in the weld.

(b) For other specimens and conditions, failure in the weld shall be acceptable, provided the stress at failure is not less than 95 percent of the ultimate strength of the base metal determined in accordance with 7.4.4.

7.5.2 Strip test coupons.- Provided that no defects associated with faulty welding conditions are present, the results of tests on coupons shall be acceptable when:

(a) The average strength of all the coupons from each specimen is not less than 95 percent of the average strength of the parent material.

(b) The minimum test value shall be not less than 80 percent of the average strength of all the coupons, except that one coupon may have a test value below 80 percent but above 60 percent provided the previous history of the tests for the particular setting shows that the frequency of occurrence of the lower test values is not greater than 1 percent.

7.5.3 Proof tests.- All specimens shall pass the specified proof test except when other methods of production testing have been approved by the procuring activity. Any noticeable yielding at the joint (determined visually) during proof testing shall be cause for rejection. If any specimen or part fails to pass the proof test, the entire production run of parts which the sample represents shall be subjected to proof testing and the schedule shall be rechecked or recertified.

7.5.4 All equipment shall be inspected periodically as recommended by the manufacturer. Adequate maintenance shall be furnished, and defective parts affecting machine operation shall be replaced before production welding is resumed.

7.5.5 Records.- The contractor shall keep a record of all tests and a list of all parts corresponding to each setup and test. The record shall include also the machine settings and other data. The records shall be subject to examination by the inspector.

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8. NOTES

8.1 Intended use.- The procedures covered by this specification are intended for aircraft and missile application, and other applications where control procedures are desired.

8.2 Definitions.- The definitions of the terms herein are, in general, in accordance with the American Welding Society's recommended practice for the flash welding of low and medium forging strength steels.

8.2.1 Weld failure.- "Weld failure" as used herein is a failure, either partially or wholly in the weld plane, except for partial weld-plane failures in heat treated whole specimens. Partial weld-plane failure in heat treated whole specimens is not considered a weld failure where fracture in the weld plane does not exceed 25 percent of the weld area and is accompanied by visual necking down of the parent metal.

8.2.2 Production run.- A "production run" as referred to herein is the production flash welding of identical parts without a change in machine setup for 4 hours of continuous operation or as established by the contractor from results of tests or prior production experience.

8.3 Protective atmosphere welding.- For greater assurance of weld quality, tubing with wall thickness over 0.149 inch should be welded in a protective atmosphere.

8.4 Figures 3, 4,and 5 are examples of initial curves established for a machine.

8.5 Publications.- When requesting publications, refer to both title and number.

Custodians:
Army - MR
Navy - AS
Air Force - 11

Reviewer activities:
Army - MI
Navy - AS

User activities:
Army - AT, ME
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Preparing activity:
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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